ST. CLARET COLLEGE (AUTONOMOUS), BENGALURU
Department of Bachelor of Computer Applications (BCA)

Vacation Mini Project Report

InsightFlow AI

Client-Side CSV Analytics & Business Insight Dashboard

Transform Raw Data Into Business Intelligence.

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Semester	**4th Semester**
Submission	**July 2026**

https://insight-flow-sales-analyzer.netlify.app/

CERTIFICATE

This is to certify that the mini project entitled

INSIGHTFLOW AI — CSV ANALYTICS & BUSINESS INSIGHT DASHBOARD

has been successfully developed and submitted by **SHRUTHI** (Roll No. **24BCA110**), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru.

This mini project was developed independently as part of self-learning during the vacation period. The work is original and has not been submitted elsewhere for any academic examination or award.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, **SHRUTHI**, Roll No. **24BCA110**, 4th Semester, BCA, St. Claret College (Autonomous), Bengaluru, declare that the mini project **“INSIGHTFLOW AI — CSV ANALYTICS & BUSINESS INSIGHT DASHBOARD”** is a record of original work done by me during the vacation period.

I declare that:

- This project was carried out independently, without any formal institutional guidance.
- The work is genuine and has not been copied from any source without acknowledgement.
- This project has not been submitted for any degree or academic title elsewhere.
- All references and third-party libraries used have been duly cited and acknowledged.
- The conclusions drawn are based entirely on my own understanding and research.

Place: Bengaluru

Date: July 2026

Signature of the Student

SHRUTHI

Roll No: 24BCA110 | 4th Semester, BCA

ACKNOWLEDGEMENT

I thank all those who encouraged and supported me throughout the development of this project.

I am grateful to the developers of PapaParse, Chart.js, jsPDF, and html2canvas — the open-source libraries that form the technical backbone of InsightFlow AI. Their well-documented, free-to-use code made it possible to implement robust CSV parsing, interactive data visualisation, and PDF report generation entirely on the client side.

I am also grateful to the broader open-source developer community and educational resources available through MDN Web Docs, Stack Overflow, and YouTube, which guided my understanding of data parsing algorithms, statistical computation in JavaScript, and dashboard UI design.

I thank my family for their constant support and encouragement throughout the vacation period.

SHRUTHI

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ABSTRACT

Small businesses, students, and analysts frequently work with sales data stored in CSV files but lack quick, accessible tools to transform that raw data into meaningful business intelligence. Professional BI tools such as Power BI or Tableau offer powerful capabilities but require installation, licensing, or cloud account setup, creating a barrier for quick, one-off analysis. There is a clear need for a zero-setup, privacy-respecting analytics tool that works directly in the browser.

InsightFlow AI is a standalone, single-file frontend analytics dashboard that allows users to upload any sales-style CSV file and instantly receive a comprehensive business intelligence report. The application parses the CSV, automatically detects its schema, computes key performance indicators, renders interactive charts, generates AI-style business insights across four categories, and produces export-ready CSV and PDF reports — all without any backend server.

Key Features:

- Drag & drop or browse CSV upload, with a built-in demo dataset for instant testing.
- Automatic CSV schema detection: numeric, categorical, and date column classification.
- Data Overview panel: total records, total columns, missing values, and a data quality score.
- KPI summary cards: Total Revenue, Avg Revenue, Max/Min Revenue, Profit Margin, Total Profit, Units Sold, Top Category, Top Region, and MoM Growth.
- Visual Analytics: Revenue by Product bar chart, Product Distribution doughnut chart, and Revenue Trend Over Time line chart, all powered by Chart.js.
- AI Business Insights across four tabs: Key Findings, Opportunities, Risks, and Recommendations.
- Business Analyst Report: an executive summary combining key metrics, trend analysis, opportunities, risks, and numbered strategic recommendations.
- Data Explorer: a searchable, sortable, paginated table of all records.
- Data Quality Report: overall quality score, missing values, duplicate rows, completeness percentage, and per-column completeness status.
- Export filtered results to CSV and generate a full PDF business report using jsPDF and html2canvas.

Technologies Used:

HTML5, CSS3, JavaScript, Chart.js, PapaParse, jsPDF, html2canvas. Deployed on Netlify.

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INTRODUCTION

1.1 Background

Sales data is one of the most common forms of business data, typically exported as CSV files from point-of-sale systems, e-commerce platforms, or spreadsheet-based record-keeping. While the data itself often contains valuable patterns — top-performing products, regional trends, profit margins, seasonal growth — extracting these insights manually requires either spreadsheet expertise or access to dedicated business intelligence software.

Professional BI platforms such as Tableau, Power BI, or Looker Studio provide sophisticated analytics but involve licensing costs, installation, or cloud account setup — overhead that is excessive for a quick, exploratory analysis of a single CSV file. A lightweight, instant, browser-based alternative that requires nothing more than opening an HTML file addresses this specific gap.

1.2 About the Project

InsightFlow AI is a single-file (insightflow.html) client-side analytics dashboard that transforms any sales-style CSV upload into a comprehensive business intelligence report. The application performs automatic schema detection, computes a wide range of KPIs, renders multiple interactive chart types, generates AI-style business insights and recommendations, and allows full export to CSV and PDF — entirely within the browser, with no backend server.

Live at: **https://insight-flow-sales-analyzer.netlify.app/** — deployed on Netlify. The repository also includes a sample-sales-data.csv file for quick demonstration.

1.3 Recommended Data Format

InsightFlow AI is optimised for sales-style datasets with columns such as Date, Order ID, Customer Name, Product, Category, Region, Quantity, Unit Price, Revenue, Cost, and Profit. The application is designed to gracefully auto-detect other CSV shapes as well, attempting to identify date, numeric, categorical, and text fields even when headers differ from the recommended format.

1.4 Scope

- CSV upload via drag-and-drop or file browser, plus a built-in demo dataset.
- Automatic schema detection across numeric, categorical, and date columns.
- KPI computation: revenue, profit, quantity, and data quality metrics.
- Interactive Chart.js visualisations for product, category, and time-series analysis.

- AI-style business insights spanning findings, opportunities, risks, and recommendations.
- Searchable, sortable, paginated data table for raw record exploration.
- Data quality auditing with missing value and duplicate detection.
- CSV and PDF export for offline sharing and reporting.

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of InsightFlow AI:

1. To build a zero-setup CSV analytics tool: Create a single HTML file that requires no installation, server, or account — simply opening the file in a browser provides full analytics functionality.

2. To implement automatic schema detection: Design a parsing layer that automatically classifies each CSV column as numeric, categorical, or date-typed, enabling the application to work with diverse real-world datasets without manual configuration.

3. To compute comprehensive business KPIs: Calculate a full suite of metrics — total and average revenue, profit margin, units sold, top category and region, and month-over-month growth — directly from the uploaded data.

4. To deliver rich, interactive data visualisation: Render multiple chart types (bar, doughnut, line) using Chart.js to visually communicate product performance, category distribution, and revenue trends over time.

5. To generate AI-style business insights and recommendations: Implement a rule-based insight engine that analyses the dataset and produces categorised findings, opportunities, risks, and actionable, numbered recommendations in natural business language.

6. To produce a polished, exportable analyst report: Combine the executive summary, key metrics, trend analysis, and recommendations into a single Business Analyst Report view, exportable as a formatted PDF using jsPDF and html2canvas, or as raw data via CSV.

7. To ensure data quality transparency: Provide a dedicated Data Quality Report showing the overall quality score, missing value counts, duplicate row detection, and per-column completeness status, so users can trust the validity of the analysis.

EXISTING SYSTEM

3.1 Current Sales Data Analysis Methods

Businesses and students currently rely on the following approaches for analysing sales CSV data, each with significant limitations:

a) Manual Spreadsheet Analysis

Excel and Google Sheets are the most common tools for sales analysis. However, building pivot tables, charts, and KPI summaries manually is time-consuming and requires spreadsheet formula expertise. There is no automated insight generation or data quality auditing.

b) Professional BI Platforms

Tools like Power BI, Tableau, and Looker Studio offer powerful analytics capabilities but require licensing fees, software installation, or cloud account setup. The learning curve for building dashboards from scratch is steep, making them impractical for quick, one-off CSV analysis.

c) Cloud-Based Analytics SaaS Tools

Online analytics platforms require uploading data to a third-party server, raising privacy concerns for sensitive business sales data, and often require subscription fees for full functionality.

d) No AI-Style Insight Generation in Free Tools

Free spreadsheet tools provide no automated business insight generation — users must manually interpret charts and figures to identify findings, opportunities, and risks, a process that requires business analysis experience.

3.2 Limitations Summary

- Manual spreadsheet analysis is slow and requires advanced formula knowledge.
- Professional BI tools involve licensing costs and steep learning curves.
- Cloud analytics tools raise data privacy concerns for sensitive sales data.
- No free tool combines automatic schema detection, KPI computation, and AI-style insights.
- Existing tools rarely provide an integrated, exportable executive business report.

PROPOSED SYSTEM

4.1 Overview

InsightFlow AI proposes a fully client-side, single-file analytics dashboard that accepts any sales-style CSV file and instantly produces a complete business intelligence package — KPIs, charts, AI-style insights, a data quality audit, and an exportable analyst report — without any server, account, or installation requirement.

4.2 Advantages

- **Zero Setup:** A single HTML file delivers full analytics functionality. No installation, no account, no server configuration.
- **Automatic Schema Detection:** The application intelligently classifies columns as numeric, categorical, or date-typed, adapting to varied CSV structures.
- **Comprehensive KPI Coverage:** Revenue, profit, quantity, and quality metrics are computed automatically, eliminating manual formula construction.
- **AI-Style Insight Generation:** Findings, opportunities, risks, and recommendations are generated automatically in natural business language, replicating the output of a junior business analyst.
- **Complete Data Privacy:** All processing occurs in the browser via PapaParse. No CSV data is ever transmitted to any server.
- **Exportable Deliverables:** CSV export for filtered data and a polished PDF business report via jsPDF/html2canvas make InsightFlow AI's output shareable beyond the browser.

4.3 Architecture

InsightFlow AI is architected as a single self-contained HTML file (insightflow.html) embedding all CSS and JavaScript. On file upload, PapaParse parses the CSV into a structured array of row objects. A schema detection function scans each column's values to classify it as numeric, categorical, or date. A KPI computation layer aggregates the numeric columns (revenue, profit, quantity) and identifies top categories and regions. Chart.js renders the Visual Analytics section from this aggregated data. A rule-based insight engine evaluates thresholds and patterns in the data to generate the four insight categories. The Business Analyst Report assembles all of the above into a single exportable view, using jsPDF and html2canvas to render the dashboard DOM into a downloadable PDF.

MODULES AND FEATURES

InsightFlow AI is organised into seven sequential dashboard sections, navigated within a single scrolling page after CSV upload.

5.1 Landing / Hero Section

The landing section introduces InsightFlow AI with a bold headline — "Transform Raw Data Into *Business Intelligence*" — and a brief value proposition. An 'AI-Powered Business Intelligence' badge sets the tone. Two primary actions are offered: **Upload CSV** (drag-and-drop or browse) and **View Demo** (loads the built-in sample-sales-data.csv dataset). A live dashboard preview widget on the right shows sample KPI values and progress bars. Three feature highlights — 10+ Chart Types, AI Insights Engine, and PDF Export Ready — summarise the application's core capabilities.

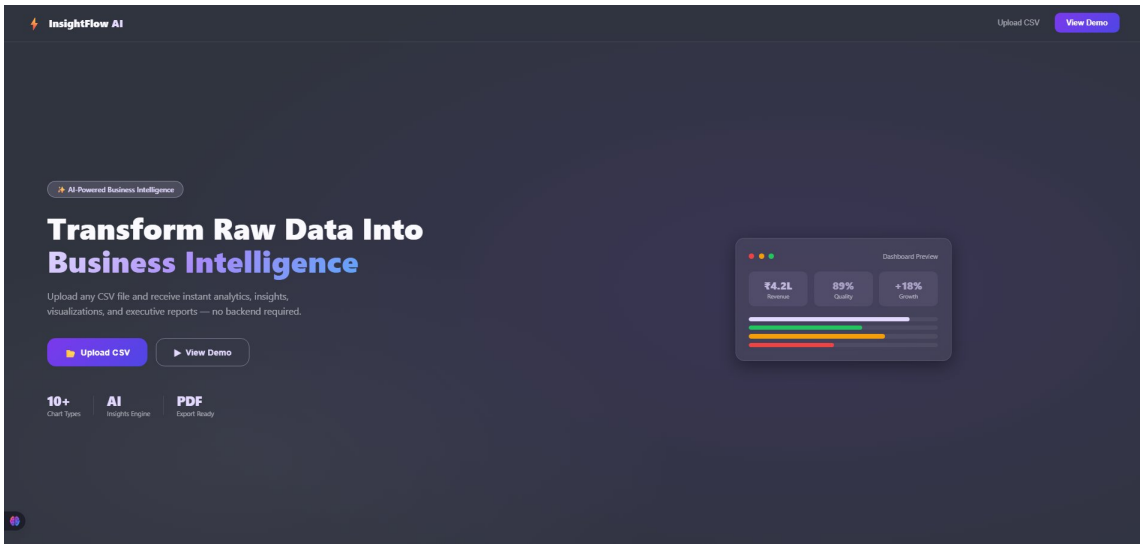


Figure 5.1 – InsightFlow AI Landing Page with Upload Options and Dashboard Preview

5.2 Data Overview and Key Performance Indicators

Upon CSV upload, the Data Overview panel displays four summary cards: Total Records, Total Columns, Missing Values (with percentage of data), and a Quality Score out of 100 with a qualitative label (e.g., Excellent).

The Key Performance Indicators section presents nine metric cards computed directly from the uploaded dataset: Total Revenue, Average Revenue per order, Max Revenue (single order), Min Revenue (single order), Profit Margin percentage with a health label, Total Profit, Units Sold with average per order, Top Category by order share, Top Region by revenue, and Month-over-Month Growth percentage.

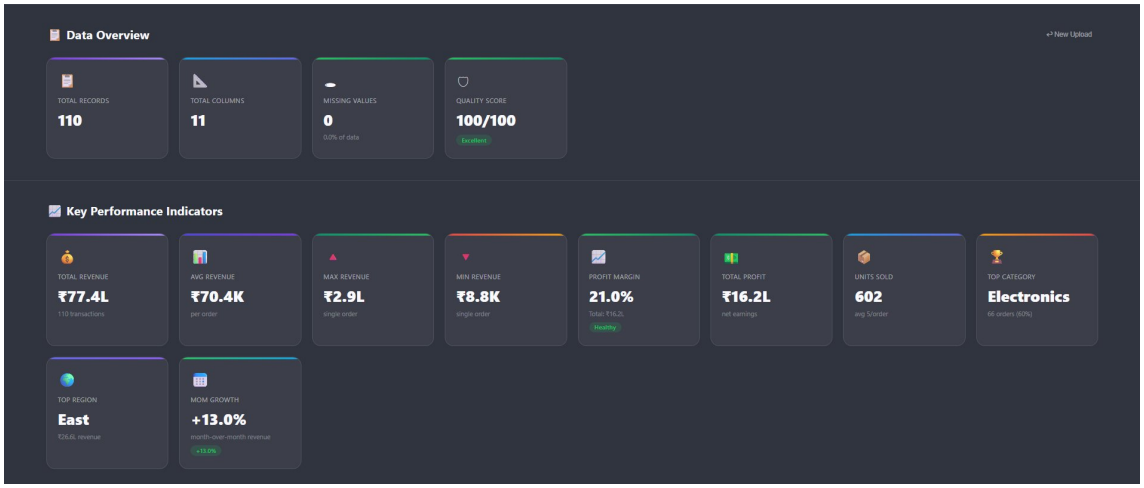


Figure 5.2 – Data Overview and Key Performance Indicators Section

5.3 Visual Analytics Module

The Visual Analytics section renders three Chart.js visualisations:

- **Revenue by Product:** A bar chart ranking all products by total revenue, with interactive tooltips showing exact values on hover.
- **Product Distribution:** A doughnut chart showing the proportional revenue share of each product, with a colour-coded legend.
- **Revenue Trend Over Time:** A line chart with an area fill, plotting total revenue across the date range present in the dataset.

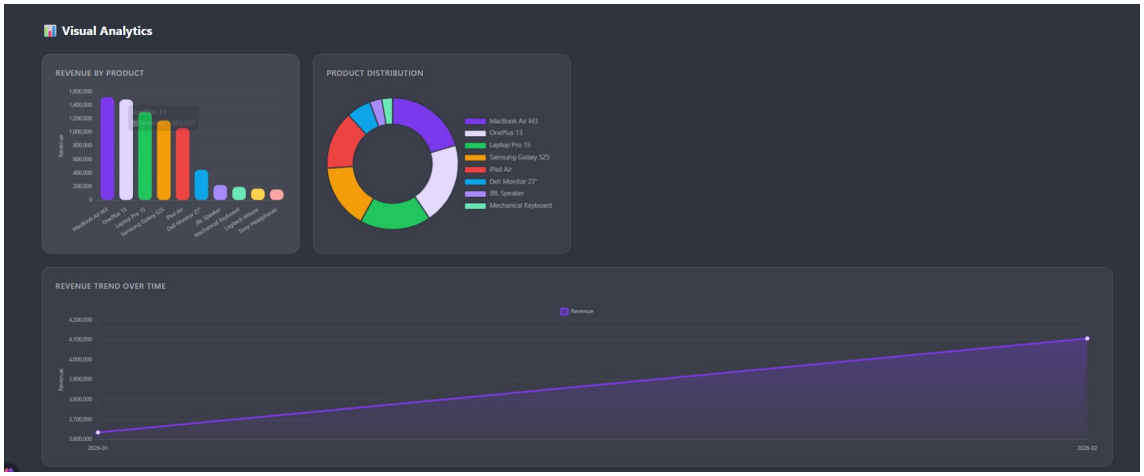


Figure 5.3 – Visual Analytics: Revenue by Product, Product Distribution, and Revenue Trend Charts

5.4 AI Business Insights Module

The AI Business Insights section presents generated insights across four tabbed categories: Key Findings, Opportunities, Risks, and Recommendations. Each insight is displayed as a card with an icon, title, descriptive text, and a category tag (e.g., Revenue, Regional, Trend, Profitability, Data, Volume).

Sample Key Findings generated from the dataset include Revenue Concentration (top 3 products accounting for the majority of revenue), Region Leader (the strongest-performing region and its contribution), Month-over-Month Growth, Profit Margin Analysis, Dataset Overview (record and column counts), and Order Volume Insights including outlier detection for unusually large orders.

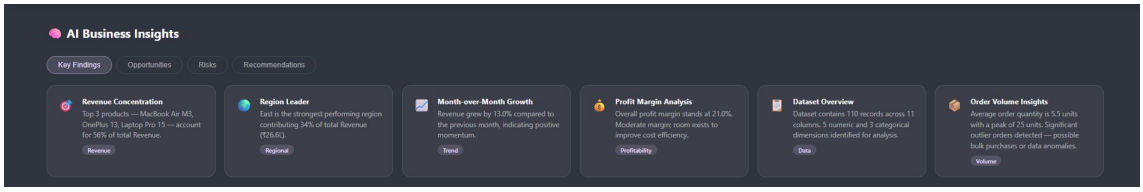


Figure 5.4 – AI Business Insights with Key Findings Tab Active

5.5 Business Analyst Report Module

The Business Analyst Report consolidates the entire analysis into a single, exportable executive document. It includes a report header (date, source filename, record count, quality score), an Executive Summary paragraph, a Key Metrics row repeating the core KPIs, a Trend & Category Analysis section listing the Key Findings as bulleted insights, an Opportunities section with specific, actionable growth suggestions (e.g., regional expansion, category rebalancing, pricing optimisation, cross-selling), a Risk Analysis section, and a numbered Recommendations section with concrete strategic actions such as doubling down on top performers, regional expansion campaigns, data quality improvement, margin optimisation programmes, and real-time analytics dashboard adoption.

Two export buttons — Export CSV and Export PDF — allow the report and underlying data to be downloaded for offline sharing with stakeholders.

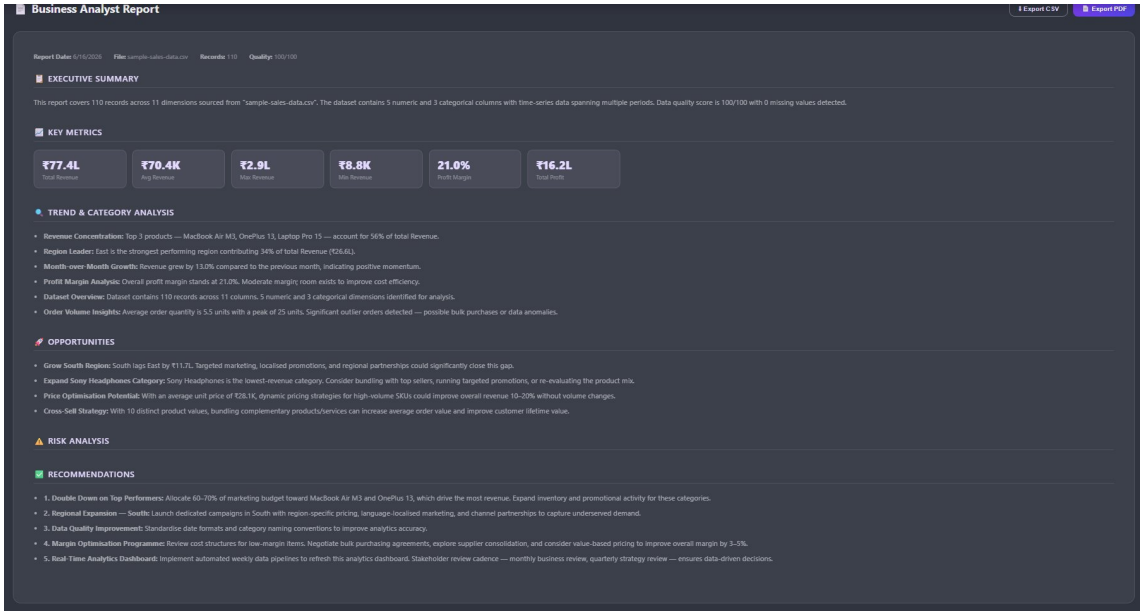


Figure 5.5 – Business Analyst Report with Executive Summary, Metrics, and Recommendations

5.6 Data Explorer Module

The Data Explorer presents the full uploaded dataset as a searchable, sortable, paginated table. Each column header includes a sort indicator, and a search box filters records in real time across all fields. Pagination controls (Prev / page numbers / Next) allow navigation through large datasets without overwhelming the browser with DOM elements.

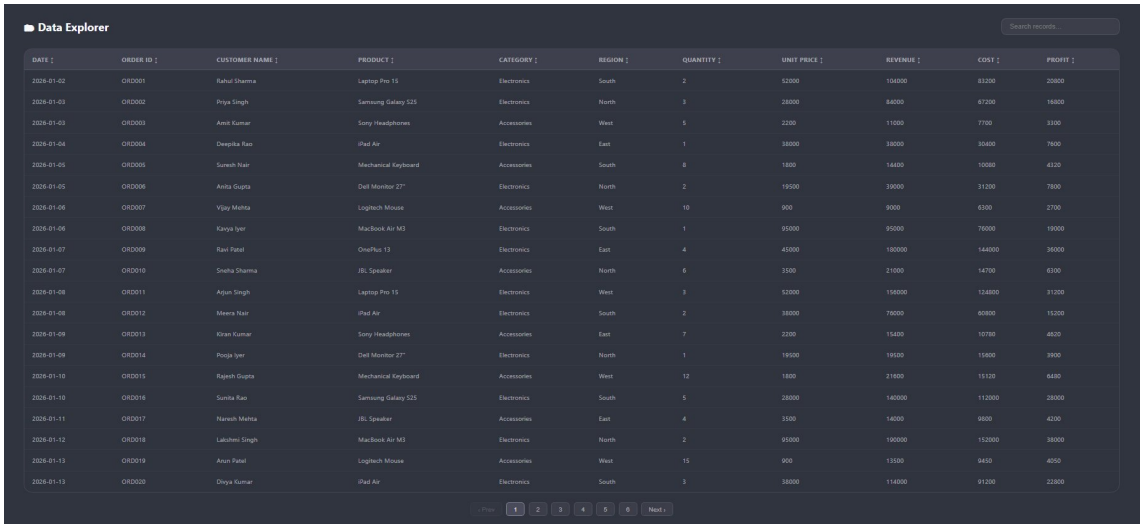


Figure 5.6 – Data Explorer with Searchable, Sortable, Paginated Table

5.7 Data Quality Report Module

The Data Quality Report provides a transparent audit of the uploaded dataset's integrity. An Overall Quality Score (e.g., 100/100) is displayed prominently with a qualitative label. A Quality Metrics panel lists total records, missing values, duplicate rows, completeness percentage, and the count of numeric versus categorical columns. A Column Completeness panel lists every detected column with a Complete or Incomplete status indicator, allowing users to immediately identify any data integrity issues before trusting the analysis.

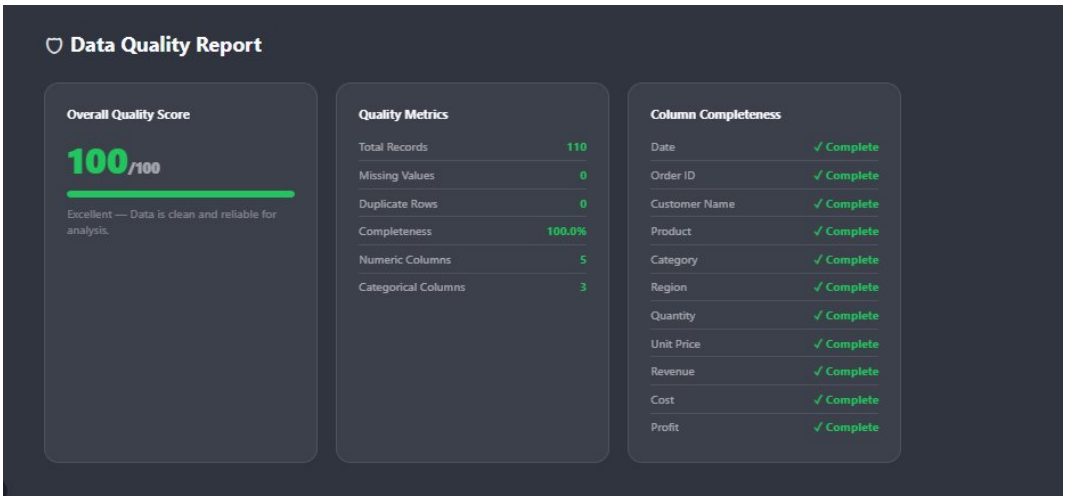


Figure 5.7 – Data Quality Report with Score, Metrics, and Column Completeness

TECHNOLOGIES USED

InsightFlow AI was developed using native front-end technologies combined with four well-established JavaScript libraries. No back-end server or database is used.

6.1 HTML5

HTML5 provides the structural foundation of the single-file application. Semantic elements organise the hero section, dashboard panels, chart containers, and the data explorer table. The drag-and-drop upload zone uses HTML5 drag event attributes, and the file browse button uses a hidden file input element.

6.2 CSS3

CSS3 delivers InsightFlow AI's dark, professional dashboard aesthetic with purple gradient accents. CSS Custom Properties manage the colour token system used consistently across KPI cards, chart containers, and insight cards. CSS Grid and Flexbox handle the responsive multi-column KPI and chart layouts. CSS gradients and box-shadows create the depth and visual hierarchy seen in the dashboard preview widget and summary cards.

6.3 JavaScript (ES6+, Vanilla)

All application logic is implemented in Vanilla JavaScript. Key implementations include: the schema detection algorithm that classifies each column based on value patterns (date regex matching, numeric parsing success rate, and cardinality for categorical detection), KPI aggregation functions computing sums, averages, max/min, and percentage growth across numeric columns, the rule-based insight generation engine that evaluates statistical thresholds (e.g., revenue concentration ratios, regional dominance, month-over-month deltas) to produce natural-language insight text, table rendering with client-side search, multi-column sort, and pagination logic, and data quality computation checking for missing values, duplicate rows, and per-column completeness.

6.4 PapaParse

PapaParse is a fast, in-browser CSV parsing library used to convert the uploaded CSV file into a structured JavaScript array of row objects. It handles edge cases such as quoted fields, varying delimiters, and large file streaming, providing the robust parsing foundation that the rest of the application builds upon.

6.5 Chart.js

Chart.js renders all visualisations in the Visual Analytics section: the Revenue by Product bar chart, the Product Distribution doughnut chart, and the Revenue Trend Over Time line chart with area fill. Charts are configured with a dark theme matching the dashboard, interactive tooltips, and responsive resizing for different screen sizes.

6.6 jsPDF & html2canvas

jsPDF generates downloadable PDF documents directly in the browser, while html2canvas captures DOM elements (such as charts and report sections) as canvas images that can be embedded into the generated PDF. Together, these two libraries power the Export PDF functionality in the Business Analyst Report module, producing a shareable, presentation-ready document without any server-side rendering.

6.7 Netlify

The application is deployed as a static site on Netlify, providing HTTPS, a custom subdomain, and global CDN distribution for the single HTML file deployment.

#	Technology	Type	Purpose
1	HTML5	Markup	Single-file structure, upload zone, table
2	CSS3	Styling	Dark dashboard theme, grid layout, gradients
3	JavaScript ES6+	Programming	Schema detection, KPI logic, insight engine
4	PapaParse	Library	Client-side CSV parsing
5	Chart.js	Library	Bar, doughnut, and line chart visualisations
6	jsPDF / html2canvas	Library	Client-side PDF report generation
7	Netlify	Platform	Static hosting and CDN deployment

CONCLUSION

InsightFlow AI is a technically ambitious and practically valuable mini project that successfully replicates the core functionality of a professional business intelligence platform within a single, dependency-light HTML file. By combining automatic CSV schema detection, comprehensive KPI computation, multi-chart visualisation, rule-based AI-style insight generation, a searchable data explorer, a data quality audit, and dual CSV/PDF export — all running entirely client-side — the project demonstrates both technical depth and genuine product utility.

The most technically significant achievement of InsightFlow AI is its insight generation engine. Translating raw statistical patterns in a dataset — revenue concentration, regional performance gaps, growth rates, profit margins, order volume anomalies — into natural, business-readable language across four distinct categories (findings, opportunities, risks, recommendations) required careful design of threshold-based rules and template-driven text generation.

The integration of four external libraries — PapaParse for parsing, Chart.js for visualisation, and jsPDF with html2canvas for PDF export — demonstrates the ability to research, evaluate, and correctly integrate third-party tools into a cohesive application, a skill directly relevant to professional front-end development.

Completing this project independently during the vacation period has been a comprehensive exercise in building a real analytics product. Designing the schema detection logic, the KPI computation layer, the insight engine, and the export pipeline provided exposure to data engineering concepts that extend well beyond typical front-end coursework.

In conclusion, InsightFlow AI is a polished, feature-complete, and publicly deployed CSV analytics dashboard that fully achieves its stated objectives and stands as a strong demonstration of data-oriented front-end development skill and independent technical initiative.

Live Project: https://insight-flow-sales-analyzer.netlify.app/